Chapter 5: The Accountability Gap

By Chris Winn

The family

One October afternoon in 2007, the journalist Katy Butler was visiting her parents in Connecticut when her mother asked her for something. “Please help me get Jeff’s pacemaker turned off,” she said. Butler’s father was upstairs, napping. He was eighty-five, a retired university professor with dementia, and the device in his chest had kept his heart beating for nearly five years. Its battery was expected to last five more.

His decline had begun with a stroke in 2001 that left him unable to finish a sentence. Butler’s mother became his caregiver, and by the next year, at seventy-seven, she was on duty more than eighty hours a week. That winter a cardiologist would not clear him for hernia surgery unless he had a pacemaker. Before the stroke he had turned the same device down on the advice of their trusted primary-care doctor, who thought it was overtreatment. Now he could no longer weigh the choice, so it fell to Butler’s mother. Exhausted from caring for him, she said yes, and the decision took minutes. The pacemaker kept his heart going while the rest of him failed. In 2006 she wrote in her journal: “My life is in ruins. This is horrible, and I have lasted for five years.”

Nothing in the way that decision was set up encouraged anyone to slow down and ask whether he should have the device at all. Their own doctor still opposed the device. He was not consulted. He was notified by fax. A visit with him could have covered what the choice meant: that the battery would last ten years, whether Butler’s father wanted to live to eighty-nine “in his nearly mute and dependent state,” and that there was another way, a temporary external pacemaker that Butler later learned could have seen him safely through the surgery. Medicare would have paid their doctor $54 for that visit, and nothing for the phone calls to the other doctors. It never happened, and nobody offered the family the other choice. What Medicare paid for was the procedure: $461 to the surgeon and about $12,000 to the hospital, an estimated $7,500 of it to the company that made the device. Butler believed her father’s doctors did their best. Her own arithmetic is blunter: “nobody makes money when medical interventions are declined.”

This was not one family's bad luck. Butler reported that a primary-care doctor like theirs netted about $173,000 a year, and a cardiologist like the one who required the pacemaker netted about $419,000. Medicare paid far more for doing a procedure than for helping a patient decide whether to have it. Dr. Ted Epperly, then board chairman of the American Academy of Family Physicians, told Butler why it stays that way: those who profit from the incentives spend money lobbying Congress and the public to keep them in place. In the year before her 2010 article, the medical sector spent $545 million on lobbying.

The surgeon and writer Atul Gawande has described how a decision like that one gets made. In his book *Being Mortal* he describes a payment system that pays well each time something is done to a patient. The same system pays poorly for palliative care, which is the care that keeps a very sick person comfortable, and it puts too little into training doctors and nurses to look after people well at the end of their lives. In a system like that, no doctor or nurse has to choose money over compassion for a family to end up where Butler's family did. Decent people respond sensibly to the incentives in front of them, and the result follows from the incentives. Those incentives did not arise by themselves. People designed them, which means people can design them differently. If they did, the same doctors would act differently, and none of them would have had to become a better or a worse person.

The company that made the device was paid. The hospital was paid, and so was the surgeon. The rest of what the pacemaker cost was paid as well, though not in money. It was paid in the years Butler's father spent in decline, and in the exhausted weeks his wife spent caring for him. Of everyone in that house, she was the person least able to refuse it. No one in that room decided to give that cost to her. So how did it get there? And is medicine the only place it lands?

The community

In Butler's family the cost was paid in her mother's hours. In many families it has to be paid in money. When a family cannot pay a medical bill, it does what families have always done, which is ask the people who know them for help. Over the 2010s more and more of that asking moved onto fundraising websites. By 2020 more than a third of the campaigns on the largest fundraising platform were medical. Between roughly 2016 and 2020 people started about 438,000 medical campaigns, and some 21 million donations brought in more than $2 billion.

That is a great deal of money, and still most of the families who asked did not get what they asked for. Only about 12 percent of those campaigns met their stated goals, which means nearly

nine in ten fell short. Roughly 16 percent received nothing at all.

The platform's chief executive has said it was never designed to pay for healthcare. It ended up doing that job by accident. It was doing the work that a working safety net would have done. Families in Canada, the United Kingdom and Germany also raise money online for medical costs. But peer-reviewed studies that compare countries find that Americans rely on it far more heavily, and that they use it in place of insurance coverage. One of those studies is the analysis by Saleh and colleagues, published in *JAMA Network Open* in 2020.

Asking for help this way also works worst for the people who started with the least. Studies find that campaigns run by people with distinctively Black or Hispanic surnames raise less money and are less likely to reach their goals, and the finding holds even after the studies account for where and when the campaigns ran. Kenworthy and Igra mapped how large these campaigns have become, and who succeeds at them, in a 2022 analysis in the *American Journal of Public Health*. How much a campaign raises depends on how many people the family knows and how much money those people have. So charity that works this way sends the least help to the people whose communities had the least to give.

In some communities the cost reaches everyone at once. In Libby, a small town in the far northwest corner of Montana, miners came home from work with dust on their clothes. It went into their kitchens and onto their children's laps. The company that owned the mine let the town use its waste as fill for the high-school running track and for gardens, so the dust spread until it was simply the air the town breathed.

The mine sat above the town. A company called W.R. Grace ran it for roughly twenty-seven years, from 1963 to 1990. The mine dug vermiculite, which is a mineral that puffs up when it is heated and becomes a light insulation that resists fire. Grace sold that insulation under the name Zonolite. For much of the time the mine was open, it produced most of the vermiculite used in the United States. It made money, and by the measures a company uses to judge itself, it was a success.

That insulation did not stay in Montana. If the house you are sitting in was built or insulated during the decades when Grace was selling Zonolite, there is a real chance some of it is above your head right now. It is a loose gray-brown fill, and it lies quietly between the joists of the attic floor. Millions of attics still hold it. So Libby is not only something that happened to a town far away. The same material may be a few feet above the chair you are sitting in.

The ore also carried a particularly lethal form of asbestos. Its fibers lodge in the lining of the lungs, and years later they produce cancers and a scarring that slowly suffocates a person. Internal company documents later released in litigation show that the contamination was identified early and that workers' exposure was measured throughout the mine's life. The mining went on.

Hundreds of Libby residents have died of asbestos-related disease, and thousands more have been diagnosed with the lung damage that marks exposure. In a town of a few thousand people, oxygen tanks rode in grocery carts beside the bread. Funerals for the relatively young were not unusual, and deaths from asbestos disease ran at many times the national rate. The Environmental Protection Agency eventually declared a public-health emergency in Libby, the first it had ever declared under the federal Superfund law.

Over its life the mine brought in revenue on the order of $100 million. That figure is an industry estimate, and no primary source has confirmed it. The federal cleanup of the town has cost more than $600 million, and in 2008 Grace agreed to pay back $250 million of that. So the cleanup alone has cost about six times what the mine took in. And the cleanup figure leaves out the deaths and the disease themselves: the years of life people lost, the hours their families spent caring for the sick, and the futures that closed. No settlement ever fully put a price on those. The figures that have been disclosed are enough to show the scale. A company took in roughly $100 million and left behind costs many times greater, and most of those costs fell on people who never saw any of the revenue.

By the time those costs were being counted, Grace had found a way out. As the claims against it mounted, the company declared bankruptcy in 2001. It reorganized, and it moved what it owed into a bankruptcy trust, where claims move slowly. In 2005 the federal government indicted the company and several of its senior executives. At the time it was the largest environmental criminal prosecution the United States had attempted. After a long trial, they were acquitted in 2009.

The acquittal doesn't mean the harm didn't happen. The death certificates, the cleanup budgets and the EPA's own emergency declaration all record it. The criminal case failed for a different reason. A prosecutor has to prove one specific act, done knowingly, at a specific moment, by a specific person. Libby's harm didn't come from one act. It came from thousands of ordinary operating decisions over twenty-seven years. Each of those decisions could be defended when it was looked at alone, and no single one of them could be proven to be the crime.

The legal scholar Katharina Pistor explains why the law works this way. In her book *The Code of Capital* she argues that capital is not something found in nature. It is made by law. It is put together out of a set of legal building blocks, which she calls modules: property, collateral, trusts, the corporate form, bankruptcy, and the choice of which jurisdiction's law applies. Those modules create wealth and protect it. They also decide who can be made to answer when something goes wrong. Seen through Pistor's work, the bankruptcy that limited what Grace owed and the acquittal that ended the prosecution were not the system breaking down. They were the system working the way it was designed to work. The same legal modules that make it possible to finance extraction on a large scale also let a corporation limit what it owes. And they let responsibility be spread across so many decisions and so many years that it cannot be pinned on any one person. Pistor shows how the law protects that separation between the people who gain and the cost they leave behind. The work of this book is to put a price on the cost that gets left behind.

Grace's shareholders and executives went on with their lives. The town stayed where it was, with the fibers in its soil and in its people's lungs.

Other communities are held together by the work they share. Along the Gulf of Mexico, fishing families had worked the same waters for generations, and parents passed on to their children the knowledge of how to read the tides and the water. On April 20, 2010, a well that was being drilled for BP in the deep water of the Gulf blew out beneath a drilling rig called the Deepwater Horizon. Eleven workers were killed. Oil poured into the Gulf for eighty-seven days before the well was finally capped, roughly 4.9 million barrels of it.

BP had drawn enormous value from the Gulf over the years, and that value was real. It was banked, and it was paid out to shareholders. When the well blew out, the cost of the spill had to fall on someone, and it fell in three ways at once. The first was a matter of place. The oil from the Gulf was refined and burned in cities far from the water, while the poison from the spill settled in the marshes where those families worked. The second was a matter of time. A quarterly earnings report looks three months ahead, and oil stays for years in a marsh and in the tissue of fish. The third was the living system itself. Toxins got into the small organisms at the base of the food web, and into the larger animals that eat them, and for years those animals reproduced less. No balance sheet was built to record what those creatures lost, because they cannot file a claim.

Here, unusually, the polluter paid, and paid more than in any other case in this book, because litigation forced it to. In 2016 BP put its own final estimate of its total cost at about $61.6 billion, and the figure passed $65 billion by 2018 as charges kept accruing. That total covers a 2015 settlement with the federal government and the Gulf states, a 2012 criminal penalty of $4 billion, the cleanup and the litigation. It still fell short. The continuing damage to the Gulf's ecosystems and fisheries is estimated at another $8 billion to $12 billion beyond what BP paid, and fifteen years after the spill those fisheries remain impaired.

Some of what the coast lost never had a price in the first place. The fishing families had handed their knowledge down from one generation to the next, and when the fishery broke, that handing down broke with it. Communities that had been organized around the water for generations lost the thing they were organized around. None of that appeared on any document BP was required to account for. Because it was never on anyone's books, it could be lost without anyone owing anything for it.

The oil did not stop at the border between the United States and Mexico. Mexico's government sued BP over the damage on its side of the Gulf. In February 2018, after five years, it settled the case for $25.5 million. Of that amount, $15.3 million repaid the government for its own monitoring and other costs, and $10.2 million went to programs to protect and restore the Gulf. Nothing was set aside for the fishing communities. The settlement also released BP from responsibility for any damage in Mexican waters. The government had paid for twenty-two scientific studies of the damage. It kept them out of its own lawsuit, and it kept them from the public. It published twenty of them only after the settlement, and only in response to freedom-of-information requests. The other two came from the government's own petroleum institute, and they were still classified when the settlement became public.

So the fishing families went to court on their own. In 2015 some 25,000 Mexican fishermen sued BP in the United States for compensatory and punitive damages. It was the first class action over the spill brought by people who were not American citizens. People on the American side of the Gulf had strong legal tools for a claim like theirs. They had the Oil Pollution Act of 1990. They had state tort law, which lets an injured person sue for damages, and state environmental law. And they had class actions, which let many people with small injuries join in a single case that a corporation has to answer. People in Mexico had far weaker tools for pressing a claim across a border against a foreign corporation. The question of how much damage reached Mexican waters was never settled, and the studies that might have answered it were held back

until the case was closed. The oil was the same on both sides of the border, and so were the chemistry and the fish. The law was different. The legal modules Pistor describes decided whether a family's loss was counted at all.

The same thing had happened twenty years earlier on another coast, and Chapter 6 gives that case its full accounting. In 1989 the oil tanker Exxon Valdez ran aground on a reef in Prince William Sound, in Alaska. It spilled millions of gallons of crude oil into one of the richest cold-water ecosystems on the continent. The legal fight that followed took most of two decades. A jury awarded a very large amount in punitive damages, and then an appeal cut the award to a fraction of what the jury had set. In the years after the spill, the Pacific herring in the sound collapsed, and it never recovered. The herring had anchored both the sound's ecosystem and the fishing economy built on it. The families who depended on the herring found no way to file a claim for the loss of an entire fishery, because the law isn't built to compensate anyone for a fishery that simply stops returning. For most of the decades since, there has been no herring season.

The country

In 2008 the distance closed. The cases so far have moved outward in rings, from one family to whole communities. In every ring, a comfortable reader has had somewhere to stand apart: it was someone else's family, or someone else's town, or someone else's coast. The financial crisis of 2008 took that place away. It struck the whole economy of the United States at once, and it reached into the retirement accounts and the homes of people who had felt safe. I was one of them.

I was doing well at the time. I worked in sales, which gave me an unusual view of the crisis, because when the shock hit every company's finances at once, companies simply stopped spending. Every expenditure that wasn't already signed got sent back for review, and I watched demand freeze from inside the sales pipeline.

I had also done what the system told someone in my position to do. I had no pension. What I had instead was a 401(k), with a number in it that had been rising slowly, and I carried the belief that came with it: with no pensions anymore, that slowly rising number was how I would one day retire. When I opened my 401(k) in the middle of the crisis, the number had been revised to essentially zero. The crash had already wiped out the rest of my investments.

The rest arrived in a pile, the way these things actually arrive. A costly medical emergency hit my finances. My car broke down, and I replaced it by tapping my retirement savings early, which meant paying the tax penalty that punishes exactly that. The company I worked for failed, and I had to pay for my own move to another city to find work. Owning a home got postponed, and retirement got pushed back by about a decade. Together, those things reset me to starting over financially.

I'm not telling this because mine was the hard case. It wasn't; plenty of people lost far more and had far less cushion to lose it from. I'm telling it because of what it takes away. This is the case in the chapter where the distance closes: the cost stops falling on people I have never met and becomes a real, almost tangible thing, because this time I had to pay it. Until now the cost has landed on someone else, somewhere else, but in 2008 it landed on millions of people who had saved. "I did everything I was told, and it reached me anyway" closes the distance more completely than "I was already struggling" ever could. The crisis didn't require anyone to gamble; it required only that they had saved, in the very instrument built for saving. So the question worth asking is not who was reckless. It is how that instrument came to be built so that a single year could erase years of ordinary prudence, and who stood on the far side of the transfer.

The 401(k) was one part of a longer change. The typical worker's pay used to rise as the economy became more productive. In the early 1970s the two came apart, as the economists Lawrence Mishel and Josh Bivens have documented at the Economic Policy Institute. The economy kept growing more productive, and the median wage, which is the wage of the worker in the exact middle, stalled. Over the same years the cost of healthcare rose at several times the rate of general inflation. So wages stayed flat while medical bills climbed, and a family with a fundraising page to pay off medical debt is living in the widening gap between the two. The political scientist Jacob Hacker gave the larger change a name in his book *The Great Risk Shift*. He describes a long, systematic transfer of economic risk off institutions and onto families. The 401(k) is one of the tools that carried out that transfer.

It began quietly in 1978, when section 401(k) of the tax code created the legal basis for a new kind of retirement saving. Over the years that followed, American employers steadily replaced the pension with the individual account. The two work differently. Under a pension, the company promised the worker a fixed income in retirement, and it was the company's job to set aside enough money to pay it. Under the new account, the worker did the saving, and the

worker lived with however it turned out. The new account was sold as freedom and ownership. Workers were told that they would control their own money, that they could change jobs without losing a pension, and that they would own capital instead of depending on a promise.

What was said less loudly was whose freedom it was. A pension left three serious risks with the company: that markets would fall, that retirees would live longer than planned, and that people would save too little or sell at the wrong moment. The new account moved all three off the corporate balance sheet and onto individuals, who cannot pool risk across thousands of lives or hedge a thirty-year horizon. The freedom that mattered most in the creation of the 401(k) was clearly the corporation's freedom from liability. For about thirty years the change caused no crisis, because nothing had yet happened to show where the risk now sat.

What finally showed it was built over a few years, roughly 2001 to 2008. The financial industry took home mortgages and turned them into securities, which are investments that can be bought and sold. Then it bundled those securities together and sliced them into more and more elaborate instruments. The complexity was no accident. Only an instrument that was too complicated to evaluate could be sold in large volume to buyers who couldn't judge what they were buying. Some of those instruments ended up in the same retirement accounts that ordinary workers had been filling with their savings since the pension was replaced. So when the housing market turned in 2008 and the instruments collapsed, the losses landed in those accounts.

The net worth of American households fell by roughly $11 trillion. That is the Federal Reserve's own count, from its flow-of-funds accounts. A single total like that hides who lost it. The economists Atif Mian and Amir Sufi worked out how the loss was spread in their book *House of Debt*. They found that it fell hardest, by a wide margin, on the households that had the least wealth and the most debt.

The reason is leverage, which means buying something mostly with borrowed money. Picture two homeowners. One owns a house outright. The other put down a tenth of the price and borrowed the rest. Now suppose the price of each house falls by a tenth. The first owner has lost a tenth of the house's value. That hurts, but it can be survived. The second owner still owes the lender everything that was borrowed, so the whole fall comes out of the down payment. The down payment was a tenth of the price, and all of it is gone. That owner has no equity left, which means no part of the house's value is theirs. The same drop in price that dents a wealthy family's balance sheet wipes out a modest one. This happened across the United States. Families

with the thinnest cushions and the heaviest mortgages lost a multiple of what wealthier families lost. They didn't behave worse. They had borrowed a larger share of the price, so the fall in price passed straight through to their equity. Mian and Sufi also show that the two groups responded very differently. For each dollar of housing wealth lost, the households with the least wealth cut their spending by about three times as much as the households with the most. When that many families cut their spending that sharply, businesses everywhere lose customers. That is how a loss concentrated on poorer, indebted households became a collapse in demand across the whole economy, and a deeper recession for everyone.

Now look at what the system could mobilize, and how fast. The Troubled Asset Relief Program committed $700 billion. When the Government Accountability Office audited the Federal Reserve's emergency lending in 2011, the cumulative total across its programs exceeded $16 trillion, and that cumulative figure is the best-sourced large number from 2008. Within roughly eighteen months, bonuses at the rescued institutions had resumed.

Households were treated differently. There were millions of foreclosure filings between 2008 and 2012. By CoreLogic's count, about 4.1 million of them ended in completed foreclosures, which means the family was put out of its home. The government found trillions of dollars within days for the institutions at the center of the crisis. For the households that were losing their homes, it organized nothing close to that in size or in speed. In 2011 the Financial Crisis Inquiry Commission concluded that the crisis had come from human action and inaction. In its judgment the crisis came from choices people made, and not from forces beyond anyone's control.

It is easy to be angry about that contrast. Being angry is not the same as being right, though, so this account has to hold up against the strongest things that can be said against it.

One objection is about the borrowers. Many of the people who defaulted could never have repaid their loans under any reasonable assumption, and they signed the papers anyway. Nobody forced them to sign. That's true. But a person can only choose well with good information, and the information around these loans was arranged to work against the borrower. The people who wrote the loans were paid according to how many loans they wrote, and not according to whether the loans were repaid. Paying people that way produces loans that should never have been written. The terms in the marketing were often different from the terms in the contract. Loans with adjustable interest rates were sold to exactly the borrowers who were least likely to understand how the rate would later reset. The Consumer Financial

Protection Bureau's later enforcement record documents this pattern in case after case. So the borrower did make a real choice, and the borrower was also given worse information than the lender had. Saying so doesn't erase the borrower's responsibility. It places that responsibility beside the larger responsibility of the people who built the imbalance on purpose.

Another objection is that the crisis was global, so it cannot be a story about American policy. European banks were heavily exposed. Asian exporters were hit hard when demand collapsed. Mortgage securities were held by investors in many countries. All of that is true, and it supports this account instead of weakening it. The instruments that spread the damage were mostly created in the United States. The practice of paying a fee for each loan written, which rewarded volume over quality, was an American design. American policy also allowed the company that issued a security to pay the agency that rated it, which is a conflict of interest at the center of the rating business. The damage spread around the world, and it started in the United States. The farther it spread, the more it matters where it started.

Another objection is that the public got its money back. TARP was largely repaid, and by some of the Treasury's measures it made a modest profit. That is also true, but it answers a different question. It tells us how the government's cash came out on one rescue program. The question here is what households lost. Families were put out of their homes. Workers saw their retirement balances reset to a lower base, and they lost the growth those savings would have compounded over the rest of their lives, and the balances never caught up. Regional economies shrank and stayed smaller. And the trust between citizens and their institutions was used up by a rescue that reached some and not others. None of that comes back when TARP is repaid. None of it appears in the accounts that the repayment argument points to.

The last objection is that reform closed the gap. The Dodd-Frank Act of 2010 was a serious response. It created the Consumer Financial Protection Bureau and the Financial Stability Oversight Council. Through the Volcker Rule, it restricted banks from trading for their own accounts. It also required banks to hold more capital and to keep more of their assets in a form they could turn into cash quickly. It is fair to read Dodd-Frank as an early and partial installment of the accountability this book argues for, and Chapter 9 returns to it as one of the first openings toward a fuller design. But a reform that is partial, and that can be reversed, has not closed the gap. The Volcker Rule was later relaxed, and the Consumer Financial Protection Bureau was later weakened. A protection that can be loosened again a few years later is not the same thing as a gap that has been closed.

All four objections rest on one belief: that in 2008 the government had no real choice, and any country would have rescued its banks and left households to manage alone. That belief is false. In the 1930s Sweden rescued its banks and its households together.

Sweden's banks were under pressure then, much like the pressure the United States faced from 1929 to 1933 and again in 2008 and 2009. In 1931 Sweden left the gold standard and let the value of its currency, the krona, fall. A cheaper krona made Swedish exports competitive again, and it made household debts lighter in real terms. Sweden also coordinated the rescue of its banks with changes to people's mortgages, so that the relief reached households and did not stop at the banks' balance sheets. Through the middle of the decade it ran public-works programs, which put income directly into households. In 1938 the Saltsjöbaden Agreement set the terms between labor and capital that the Swedish welfare state was then built on. The scholars who compare banking crises, working in the tradition of Charles Kindleberger and Barry Eichengreen, treat Sweden as standing proof that a country does not have to choose between rescuing its banks and relieving its households. Chapter 9 develops the comparison. The United States chose otherwise in 2008 and 2009, and it chose much the same way again in 2020. Nothing forced the United States onto the path it took. It chose that path, and Sweden shows that another one was open.

The United States holds one more compact, and it runs between parents and children. Social Security is the promise that each working generation will care for the one before it, so that across a whole country the children pay for the parents. It was built during the Depression so that growing old would not mean being left with nothing. It is not a fraud and not a Ponzi scheme. What follows takes aim at one feature of how its promises were funded, not at the program. The distinction matters, because a careless version of this argument is exactly the kind of thing that gets turned into a weapon against a program worth defending.

Social Security was designed in 1935 as a pay-as-you-go system. That means the payroll taxes that today's workers pay are used to pay the benefits of today's retirees. For decades the program took in more than it paid out, and the surplus was held in a trust fund, in the form of special Treasury bonds. Nobody needed to forecast when the baby boom generation would retire. Those people had already been born and counted, so the years when they would start to retire could be read off a calendar decades ahead. By the early 2020s the system was paying out more than its payroll taxes brought in. Under current law and current projections, the trust fund runs

out around 2033. After that, the payroll taxes coming in would cover roughly three-quarters of the benefits that have been promised, about 77 percent.

The size of that gap can be stated as one honest figure. The program's Trustees publish a seventy-five-year actuarial shortfall of roughly $22.6 trillion. That is the projected gap between the benefits that have been promised and the revenue expected to pay for them.

So where, in Social Security, did a cost get passed to later generations? In the way benefits were promised, and there is a year when you can watch it happen.

In 1972 Congress passed an across-the-board increase in Social Security benefits of about 20 percent. The same law created the automatic annual cost-of-living adjustment that has tied benefits to inflation ever since. The increase took effect for September 1972, so the larger checks reached beneficiaries in the first week of October, about a month before the November election. The payroll-tax increase meant to pay for them didn't take effect until after the election. The reward reached voters before they voted. The cost was scheduled to arrive after they had.

The 1972 law also had a flaw in the method it used to index benefits to inflation, and the flaw raised benefits by more than Congress had intended. Congress corrected it in 1977 by raising taxes and trimming benefits. By the early 1980s the program was in trouble again. Estimates showed the retirement trust fund running out as early as August 1983, and that spring Congress passed another rescue, which raised taxes and cut benefits once more. Each time, the bill fell on whoever happened to be working, paying in, or about to retire when it came due.

The record makes the case, and nothing has to be added to it. Nobody has to claim that someone in 1972 privately worked out the deficit of the 2030s and hid it. The case needs only what the record plainly shows. A benefit increase was timed to arrive before an election, and the tax to pay for it was timed to arrive after. The increase was passed at a time when there were many workers for each retiree. That this ratio would eventually fall was public arithmetic, not secret foresight. The people who made the promise collected the political reward for it while it was cheap to make, and they were out of office before the bill fully arrived. Paying for it fell to the workers of later decades. Many of them had not yet been born, and none of them could refuse.

Social Security did not have to be built this way. The 1935 design was one option among several. Chapter 4 described Norway's sovereign wealth fund. Norway takes the rents from its oil and

gas, meaning the income they bring in beyond the cost of producing them, and puts that money into a fund that compounds for future generations, instead of spending it as it arrives. The United States and Norway faced the same question, which is what one generation owes the next. They answered it with different designs, and they got different results. The difference in results is not explained by national wealth or by demography. It is explained by a choice that specific people made at a specific moment. A design that set money aside in advance was a real alternative. So the pay-as-you-go design was selected. It was not fated.

Two objections deserve a hearing. The first comes from public finance, and it is entirely correct. It says that holding the program's reserves in Treasury bonds is what a sound fund manager does, and that putting retirees' basic security into the stock market would have added exactly the kind of ups and downs a retirement system exists to remove. That is right. But it defends something this argument does not criticize. The argument is not about how the surpluses were invested. It is about how the promises were made. A defense of the investments, however sound, says nothing about the timing of the promises and of the taxes that were supposed to pay for them.

The second objection says the gap can be closed. Congress could raise the payroll-tax rate. It could lift the cap on the wages that are taxed. It could trim benefits at the top, or raise the age of eligibility as people live longer. The arithmetic is right. The gap can be closed, and closing it is the responsible thing to do. But look at who pays under every item on that list. Each one puts the obligation on the people who will be living through the years of deficit: the workers who will pay a higher rate, receive a smaller benefit, or work extra years. None of it reaches the generation that collected the political reward for making the promise, and most of that generation is now beyond the reach of any adjustment. So a gap that can be closed is still a cost that was separated from the people who created it. It is arriving on the schedule the calendar set decades ago. The fact that the bill can be paid doesn't change who is handed it.

The world

In the villages near Baotou, in Inner Mongolia, people report raised rates of cancer, respiratory disease and birth defects. Farmland around the processing plants has gone out of use, because the soil can no longer grow food.

The plants refine rare earths. These are the elements used to make the strong magnets in electric motors and wind turbines. The ore comes from the Bayan Obo deposit nearby. This one

place produces roughly half of China's output of rare earths, and China supplies on the order of 60 percent of the world's.

The finished products are clean, and the refining is not. To separate the valuable elements from the ore, the plants use large volumes of strong acid. The process leaves behind enormous amounts of toxic waste that contains heavy metals and leftover acid. The ores are often bound up with thorium and other radioactive elements, so the waste is mildly radioactive as well. A conservative accounting puts the wastewater at roughly ten tons for every ton of finished rare-earth oxide, and that is the low end of the published estimates. The waste flows into vast holding ponds, called tailings ponds.

At Baotou the cost is separated from the people who benefit in every way a cost can be. It is separated in time. The ore is paid for today, and the damage to people's health and to the land plays out over generations. For the radioactive waste, it plays out over something closer to geological time. It is also separated by place and by power. The value goes to shareholders, and to consumers in rich countries with strong institutions. The cost settles on rural communities that have little political voice and few ways to refuse.

The evidence from this far away is thinner, because China's information environment restricts what can be gathered and published about health and the environment around Baotou. Peer-reviewed work exists all the same. Zhang and colleagues documented the environmental costs of the rare-earth industry in 2022, and Zhou and Ge did similar work in 2021, both recording how much waste the industry makes and the burden it puts on the region. Nobody can give a tidy total for the cleanup. A figure of about $100 billion sometimes circulates, but it doesn't trace to any source that survives checking, so it stays out. What can be said is narrower. The industry's annual environmental cost has been estimated in the billions, its revenue runs on the order of $10 billion to $12 billion a year, and the share of that revenue spent on remediation has historically been a small fraction of the harm.

Rare earths also show a limit on what any fund can do. They are the clearest example in this book of a resource that has almost no substitute. When a country draws down its oil, it can at least invest the rents for the future, the way Norway's fund does, so that something is built up to make up for what is used up. Nothing like that is possible with rare earths. No fund can rebuild a deposit once it is depleted, and nothing that can be engineered can stand in for these elements at the scale modern technology needs them. When substitutes are that scarce, the cost can only be prevented or paid. Reinvesting the proceeds cannot make it good. In the appendix

to this book, the limit of substitution is a parameter in a formula. At Baotou the limit of substitution stops being a parameter in an appendix and becomes a field of tailings ponds.

In your pocket, and in mine

Now think of the phone in your pocket, and the one in mine. Rare earths go into their speakers and their screens. They also go into the motor of an electric car and the generator of a wind turbine. In wealthy economies, people treat the devices built from rare earths as ordinary, and more and more as clean. The tailings ponds are in Inner Mongolia. The engineer who designs the phone does not see them, and neither does the household that charges it.

That doesn't make you or me a villain. I am not saying that to soften the case. It is the center of the case. You and I share in the value the extraction produced, and the whole length of the supply chain lies between us and its cost. Whether either of us is a good person, or feels guilty, has nothing to do with it. The most virtuous person alive could own one of these phones, and the ponds would be exactly the same. The distance that a global supply chain creates is not an unfortunate side effect. The distance is what makes the arrangement livable for the people who benefit, because it keeps the people who pay in a place where the people who gain will never see them. Nobody has to be cruel for this to work. In this case the cost only has to stay out of sight, and a supply chain that circles the planet keeps it there.

With the phone in your hand, look back through the rings. In Butler's family the person who paid was in the same house. In Libby the people who paid were in the kitchens where the dust came home. In the Gulf they were out on the water, on both sides of the border. In 2008 I was one of them. In Social Security many of them are not yet old enough to work, and at Baotou they live in villages most of us will never see. In every ring before this one, the reader has stood near the people who pay, and sometimes among them. The phone shows the other side. In some ring of every life, each of us is also among the people who gain, and nobody has to choose that for it to happen. That answers the question from Butler's house, which was how a cost reaches the person least able to refuse it when no one in the room decided to send it there. The people who gain are kept apart from the people who pay, and the cost goes to whoever cannot refuse it. The same arrangement runs through every ring, and it runs through us.

The pattern, named

Put these cases next to each other and they have the same structure. Something shared was drawn on as though it cost nothing: a mineral seam, a coastline, a fishery, a supply chain, a nation's savings, a promise between generations, a dying body.

It works like this. A company, a generation or a government draws value from something it depends on but did not produce. That might be a mineral seam, a fishery, a river or the air. It might be the labor of other people, the trust between citizens, or the future itself. Drawing that value creates a cost. The cost is then cut loose from the people who benefited, and it is pushed onto people who have less power, less ability to move away, and less ability to refuse.

The cost comes due later. By then the mine has closed. The company has reorganized. The executives have cashed out. The politicians who made the promises have left office. The cost stays with whoever was left standing where it landed.

The one reliable thing about this pattern is that the bill comes due. It is never cancelled. It is only put off and handed to someone else. So the only open question is who ends up holding it.

What you have been watching in every ring has a name. It is a severed cost: a cost cut loose from the party that drew the value and left with whoever could not refuse it. The weeks Butler's mother spent on duty were a severed cost. So were the lungs of Libby, the Gulf's lost seasons, the savings erased in 2008 and the farmland around Baotou. Once you can name it, you start to see it everywhere.

The word has a second meaning that could mislead. Severance is also the name for the check a company writes when it lets a worker go. A reader who hears that meaning first might suspect that the term treats labor as a commodity, which is a view this book resists. It does the opposite. Both meanings describe a cut. In one, the worker is cut loose from the wage. In the other, the cost is cut loose from the party that caused it. I keep the word because the cut is the thing I am describing. The accounting already makes the cut disappear from the books, and a gentler word would make it disappear from the language as well.

How does the party that gains slip the bill? Chapter 7 names the full set of covers, and a few are enough here. Distance keeps the tailings at Baotou half a world from the people charging their phones. Time let a Congress in 1972 collect the reward for a promise whose bill falls due in the 2030s. A gap in knowledge separated the company that measured the asbestos from the families

who breathed it. A gap in who can leave means capital moves and a poisoned town cannot. Routine lets a dying fishery or a family's bankruptcy fade into background noise. And abundance makes a real cost look like none at all, at least for a while: cheap insulation, cheap electronics, a rising retirement balance.

Then consider who absorbed the cost in each case. A town that could not move. Fishing communities that could not relocate. Villages with no way out. Households that were foreclosed on, and future taxpayers who cannot opt out of what they will inherit. Families whose fundraising pages fall short, and a woman whose eighty-hour weeks of caregiving appear on no ledger. In every case the party that gained was able to leave: the company went bankrupt, the politicians left office, the generation that made the promise grew old. And in every case the cost stayed with the people who could not get away from it.

These cases don't show a system failing. They show a system doing what it was built to do. No one has to intend harm for the pattern to appear. The rules reward whoever draws the value, and they push the cost onto whoever is least able to resist, and that is enough to produce it.

None of this is a new insight. In *The Social Costs of Private Enterprise*, published in 1950, K. William Kapp argued that shifting costs onto third parties is a routine feature of private enterprise, not an occasional failure at the edges of the market. The traditions of political ecology and environmental justice that grew up afterward documented who those third parties reliably are. In the work of scholars such as Joan Martínez-Alier and Robert Bullard the answer is consistent: the poor, the rural and the politically voiceless. The diagnosis is theirs.

Some existing tools come close to what this book adds, without quite reaching it. After Deepwater Horizon, regulators assessed the damage to the Gulf with a method called natural-resource damage assessment. That method works only after the harm is done. In 1990 Robert Costanza and Charles Perrings proposed something earlier: requiring a bond, which is money set aside as a guarantee, to be posted in advance. That is the right timing. But they set the amount of the bond against the worst damage that could possibly occur, and not against the specific cost that a given extraction severs. That specific cost, priced for a single extraction, is what this framework sets out to measure.

Kapp and those who followed him showed who is left holding the bill. What this book adds is the amount written on it, posted as a reserve before the bill comes due, or to pay a bill long overdue.

One kind of thing

Ask what each of these cases drew on. A reef, the atmosphere, a community of fishing families, a body of human labor and a generation not yet born look like entirely different kinds of things. Define a commons by its structure rather than its substance, though, and they become one kind of thing. A commons is anything people depend on and share that was never produced as a commodity for sale. Nobody manufactured a reef. It was inherited. Nobody built the atmosphere to a specification. A community's knowledge and trust were never made to be sold, the care that raises the next generation was never produced as a market good, and a future generation is the purest case of something inherited rather than made.

By that definition the reef, the atmosphere, the community and the unborn are the same kind of object. Each one is a commons, and each was so abundant that drawing on it seemed to cost nothing, when in fact it carried a cost. That is why one method can price all of them. The method works in two directions. It prices forward, against a future that is being foreclosed. And it prices backward, against extraction from communities that has already happened.

Putting a price on any of these invites a serious objection. The philosopher Michael Sandel has made it as well as anyone, in his book *What Money Can't Buy*. Sandel warns that when you put a price on a sacred landscape, a community or the unborn, you turn something priceless into a transaction. You accept the extractor's way of seeing, in which everything can be exchanged for everything else and everything has a number. Worse, a price can make a harm legitimate. It turns a violation into a fee, something you are allowed to do as long as you pay. That worry has to be answered.

The answer depends on what really happens when these things are left unpriced. If the choice were between a sacred thing with no number and a sacred thing reduced to a number, Sandel's worry would settle it. That is not the choice. These things already have a number, and the number is zero. Zero is what the extractive accounting assigned to the reef, the community and the future. Zero is what made the extraction profitable. When destroying a reef costs nothing on the books, the extraction that destroys it shows a profit, and that profit is how it got financed.

This book refuses that zero. It says the cost was never zero, and it sets out to say what the cost is. There are two different acts here that happen to share a word. One is pricing something in order to sell it. The other is pricing something in order to hold a reserve against its loss. Pricing

to sell turns a forest into board feet of lumber. Pricing to reserve states what the forest is worth, so that destroying it can no longer be recorded as free.

That refusal has an ancestor. In 1944 Karl Polanyi published *The Great Transformation*. The book is built around an idea he called the fictitious commodity. Polanyi argued that the market treats labor, land and money as if they had been produced for sale, when none of them was. Labor is human life. Land is nature. Money is a token of trust among people. Running a society as though these were ordinary goods does violence to them, and to the people and places they actually are.

This book's definition of a commons descends from that thesis, and so does the chapter's basic test: imagine the commons gone and ask what cost appears. Polanyi's deepest idea was what he called the double movement. When something a society depends on is turned into a market good, the society pushes back, and it ties that thing to social obligation again. Polanyi described the double movement as a tendency he could see in history. This book turns it into a working instrument.

The living thinker closest to this argument is Nancy Fraser, and her work settles a debt this book took on several chapters ago. Fraser distinguishes exploitation from expropriation. Exploitation takes from workers through the wage. There is a contract, however unequal it is. Expropriation takes without even the form of a contract. Fraser has two things in mind. One is the unpaid care work that keeps a society going. The other is nature: its resources, and its sinks, which are the places that absorb waste. Capital relies on all of these. They sit outside the wage relation, and they are simply taken. Fraser lays this out in her essay "Behind Marx's Hidden Abode" and in her book *Cannibal Capitalism*.

The reef, the community's knowledge and the caregiver's labor were all taken without a contract. They are expropriation in exactly Fraser's sense. So are the options of a future generation, which can never be a party to any contract at all. Chapter 2 traced the word severance and left one thread unfinished, and this is where it is finished. The name that thread was waiting for is Fraser's.

The harshest reading of this whole project also needs an answer. A critic might say that pricing the commons is financialization by another route, in which the commons gets repackaged as a new class of asset for someone to trade. Polanyi and Fraser would not endorse putting a price on nature, and this book doesn't claim they would. Their tradition refused to treat nature as an

asset. This framework stands inside that refusal, because accounting for a cost is a different act from creating an asset to trade.

The most forceful living version of this worry comes from Brett Christophers. His work traces how a financialized economy puts the wrong price on whatever it turns into a tradable asset. A later chapter takes up his argument in full. The answer in principle comes from a test that Katharina Pistor supplies.

In *The Code of Capital* Pistor names four legal attributes that turn an ordinary asset into capital. Priority means a claim ranks ahead of other claims. Durability means it keeps that rank over time. Universality means it binds everyone. Convertibility means its holder can turn it back into state money. An accountability bond is built on purpose to lack the two attributes that let capital circulate. It is not convertible. It can't be redeemed for money, or for anything except the discharge of the specific liability it records. And it is not transferable. It can't be sold, traded or repackaged into a security. A debt that can only be paid, and can never be traded, is the opposite of what Pistor means by a coded asset. That is the difference between creating an asset and accounting for a cost, stated in the terms of Pistor's own test.

Who shaped the system?

The ledger shows who ended up with the cost. It doesn't show who wrote the rules that sent the cost to them. This book's method accounts for who bore the cost. Beside that sits a separate question: who shaped the system so that the cost would go where it went?

A rigorous set of tools for that question already exists, and there is no good reason not to use it. James Buchanan and Gordon Tullock built the tradition known as Public Choice to analyze how concentrated interests shape institutions to serve themselves, while the cost is spread thinly across everyone else. At its center is Tullock's analysis of rent-seeking. Rent-seeking means spending real resources to capture a privilege, instead of spending them to create something of value. Mancur Olson described the imbalance underneath it in 1965. A small group with a great deal at stake organizes. A large group in which each person has only a little at stake does not. So over time the rules drift toward whoever shows up.

The analysis runs cleanly where the people who take the value and the people who bear the cost are different populations, and in most of these cases they are. First identify the interests that shaped the institutions. Then trace the lobbying and the coalitions behind the design. Then

ask whom the design serves and whom it costs. When those steps are applied to the cases in this chapter, a list of names appears.

At Libby, internal company documents released in litigation show lobbying and an effort to delay the regulation of asbestos. In the Gulf, offshore drilling was overseen by a federal agency called the Minerals Management Service. It is a textbook case of a captured regulator, which means a regulator that has come to serve the industry it is supposed to oversee. Its staff had accepted gifts and travel from the industry, and people moved back and forth between the agency and the companies it was supposed to oversee. The Interior Department's own inspector general had reported on the capture before the disaster, and after the blowout the agency was broken up and reorganized.

The deregulation that made 2008 possible came in three main pieces. The Gramm-Leach-Bliley Act of 1999 took down the wall between commercial and investment banking. The Commodity Futures Modernization Act of 2000 exempted derivatives from regulation. A series of federal rulings pushed aside state consumer protections. Financial-industry lobbying shaped all three, as the Center for Responsive Politics and congressional hearings documented at the time.

In healthcare, the pharmaceutical industry has consistently led American industries in lobbying spending, and for decades Medicare was barred from negotiating the prices of the drugs it paid for. That ban was loosened only recently. It is as clean a rent-seeking outcome as the literature contains. Chapter 8 adds the coal economy of McDowell County, West Virginia, where an industry spent generations shaping a state's laws and regulations to its own advantage.

The analysis doesn't run cleanly everywhere. At Baotou, the demand of consumers around the world shapes the extraction as much as any lobby does, and it is hard to separate one actor from another. In Social Security, the people who will bear the future cost were partly the people who paid in during an earlier period, so what happened across time cannot be reduced to a list of interests. Where the analysis blurs like that, the account of who bore the cost has to carry the argument alone. It can, because it doesn't depend on finding a lobby.

It is easy to misread how I am using these tools. I draw on the vocabulary of rent-seeking without endorsing Buchanan's larger political project, much of which points where this book doesn't go. A tool can be used by people who disagree about nearly everything else. The two kinds of analysis also need each other. Public Choice asks who shaped the rules, and to finish

that analysis it needs a ledger of who bore the cost, which is what this book builds. Neither one replaces the other. Readers who want the rule-shaping side in full should read Buchanan, Tullock and their successors directly. They will find that the vocabulary of rent-seeking holds up under many different readings of Buchanan's broader commitments.

Chapter 9 develops this pairing at length. It is meant as an invitation to readers at institutions like the American Enterprise Institute, Cato, Mercatus and George Mason University, where the Public Choice tradition is most at home. The ledger this book builds is the natural complement to the rent-seeking analysis they already do, and each is stronger with the other.

Restitution and foreclosure

If all of this is one mechanism, what would honest accounting for it require? The framework's answer is a pair of instruments. Chapter 6 supplies the method for them, and Chapter 9 takes up their politics.

The gap looks different depending on which way in time you face. Facing backward, there is harm already done: the lungs of Libby, the marshes of the Gulf, the herring of Prince William Sound, the households foreclosed on in 2008. Facing forward, there is harm not yet done. That is the cost that extraction today is imposing on people who don't yet exist, as a habitable atmosphere is spent, irreplaceable materials are drawn down, and options close. A design that answers only one of the two answers half the problem.

So the framework proposes two instruments. The Restitution Bond discharges harm already done. It looks backward, and it is owed to particular people and communities who exist and have been hurt. The Foreclosure Bond discharges the cost current extraction imposes on the future. It looks forward, and it protects outcomes for people who don't exist yet. Together they make up the aggregate accountability bond the framework requires. Each is measured against its own side of the cost: the Restitution Bond against damage already done, and the Foreclosure Bond against the remaining value of the commons minus whatever has already been posted to protect it.

Why bonds, and not a tax, a fee or a fund? A tax goes into general revenue, and nothing ties the amount collected to a specific future obligation. A fee buys permission to go ahead, so it is a toll and not a remedy. A fund pools money without paying off any particular liability. A bond differs from all three. It is posted before the full cost has been borne. It stays in place across the

many years over which that cost arrives. And it pays off something you can observe: a specific posted amount against a specific identified liability.

Bonds of this kind already exist, which is part of why the word is honest. Mining companies post reclamation bonds under the Surface Mining Control and Reclamation Act of 1977, against the cost of restoring the land. Oil and gas companies post decommissioning bonds against the future cost of closing a well. Norway's sovereign fund works like a bond at the scale of a nation, holding capital against future need. The economics of reparations treats a historical injustice as a documented debt to be discharged. And the civic-republican tradition holds that the living have an obligation to future generations, and a bond is the natural way to put that obligation on the books. The word bond gathers all of these together.

The backward instrument is called Restitution, and the name places it inside a tradition, not beside it. William Darity Jr. and A. Kirsten Mullen define reparations as "a program of acknowledgment, redress, and closure for a grievous injustice." In a 2024 essay with Thomas Craemer, Daina Ramey Berry and Dania V. Francis, Darity spells out the middle term: "Redress is the act of restitution, the specific steps taken by the culpable party or their successors to provide compensation for damages to the victims or their heirs." The Restitution Bond is built to carry out that act. It is not a rival to their framework or a new name for it. It is an instrument their framework already has room for.

The name of the forward-looking instrument does similar work. A reader who hears the word "foreclosure" probably thinks first of 2008, and of a family barred from its home. That connection is intended, and there is nothing in it to apologize for. The word comes from the old *forclore*, which means to bar from, and it describes both events exactly. A mortgage foreclosure bars a family from its home, through a set of laws that lets the barring happen on a large scale. The Foreclosure Bond addresses future generations who are being barred from what they should inherit: a habitable atmosphere, intact ecosystems, irreplaceable materials. The mechanism is the same in both, and only the scale differs. In one case it is a household, and in the other it is a generation.

The other candidate names miss what matters. "Future generations" reads as soft philanthropy. "Sustainability" has been absorbed into corporate branding. "Reclamation" is already a narrower legal term. Both "conservation" and "intergenerational" miss the specific act of barring that "foreclosure" names.

Now the lineage behind the two instruments can be placed. Pistor supplies the legal half of the diagnosis: she shows how law encodes and protects the severance, and the framework prices what the law severs. For the forward instrument's moral footing, Derek Parfit's *Reasons and Persons* (1984) supplies the standing of future people. It offers a way to take our obligations to people not yet born seriously at the level of outcomes rather than particular identities. The framework leans on Parfit without claiming to have solved the non-identity problem he famously left open, and it names that problem as a standing limit rather than treating it as defeated.

Two more traditions stand behind the instruments. One is civic republicanism, as set out in Philip Pettit's *Republicanism* (1997) and Quentin Skinner's *Liberty Before Liberalism* (1998). It treats freedom as non-domination, and it does its strongest work on harms within a single generation, like the ones this chapter documents. The other is John Hartwick's rule of 1977. The rule says that the rents from a resource that is being depleted should be reinvested, so that later generations are no worse off. It is the ancestor of the forward-looking instrument, and Norway's fund is its cleanest application.

The backward-looking instrument descends from the reparations tradition. When Ta-Nehisi Coates published "The Case for Reparations" in *The Atlantic* in 2014, he supplied the shape of the argument: name the mechanism, name the harm, name the debt. This framework extends that shape to communities built around extraction. The economics comes from Darity and Mullen's *From Here to Equality* (2020), which is the field's contemporary landmark and this framework's primary source. It also comes from *Umbrellas Don't Make It Rain* (2015), by Darrick Hamilton, William Darity Jr., Anne Price, Vishnu Sridharan and Rebecca Tippett. They compared households that were matched on other characteristics, and they showed that the structural variable is wealth, not behavior: Black households at the highest level of education hold less wealth than white households at lower levels. The line of work reaches back to Dalton Conley's *Being Black, Living in the Red* (1999), an earlier statement that wealth is the variable that matters.

The method belongs to them as well. For the debt owed to the descendants of people enslaved in the United States, the pricing is Darity and Mullen's work, and this book defers to it. I should say plainly why. I am not a descendant of people enslaved in this country, so I don't feel it is my place to set the figure for what they are owed. That accounting has been built by scholars who have given their working lives to it. Their measures of the racial wealth gap, and

of the gap in life expectancy, put a price on the legacy of a historical injustice, and they are the most developed backward-looking accounting in the field. When this book measures McDowell County's own gap in life expectancy, it borrows their method and applies it to a different harm. What this framework offers is a fit, not a replacement. It prices harm in both directions, and pricing backward is what their method already does. The same apparatus that can meter what an asteroid-mining colony draws from a commons no one has touched yet can carry their accounting of a debt already incurred, on their terms. Applied to that debt, the Restitution Bond is a reparations bond. Its amount is the one Darity and Mullen's method sets, and this book's part is only to show that the debt can be posted and kept on the books until it is paid.

The backward-looking instrument applies the same method of matched comparison to communities built on extraction. McDowell County's collapse in wealth can be measured against comparable Appalachian counties that never depended on coal. Libby's health costs can be measured against similar towns that never had a Grace mine. Appalachia's opioid deaths can be measured against comparable counties where Purdue's documented marketing did not concentrate, and Chapter 8 runs that third comparison in full.

A current program in climate economics reached the two-directional debt before this book did. Andrew Fanning and Jason Hickel have calculated the compensation owed for the appropriation of the atmosphere. The Caribbean Community (CARICOM) has put forward a plan for reparatory justice. The Loss and Damage Fund was established at the 2022 United Nations climate conference, and it began operating the following year. They were first, and this framework says so. The framework concedes the priority and adds only the grain: it prices a single extraction event and reserves against it, rather than netting one nation's account against another's.

One question this book does not presume to close. How should coerced labor be priced, as distinct from its legacy? Darity and Mullen price what can be measured of that legacy, and the framework follows them: wealth gaps, gaps in life expectancy, broken transmission of knowledge, trauma passed down through generations. The direct pricing of coercion belongs to the people already carrying that question forward. An honest ledger can name a line and leave it without a figure. An empty line is not a zero. It says a cost is real, that it belongs in the account, and that the amount is not this book's to enter. The dispossession of Indigenous land

and the cultural knowledge that colonization severed sit in the same territory. The method can reach them in principle, but the computing belongs to later work.

How much has actually been posted in bonds across American extraction today? Far less than the remaining value of the commons the bonds are meant to protect. The federal black-lung program is one measure. The Government Accountability Office and the Congressional Research Service put the program's payouts at roughly $44 billion through 2009, which is the most recent authoritative total. That is a real number, and it is a conservative floor for what the program has paid over its full run. Even so, the program's trust fund still carried $5.1 billion in debt as of September 2024.

The more important finding is not that the instrument is missing. It exists, and at large scale. The problem is that part of it is assigned to the wrong party. The black-lung program is paid for by an excise tax on coal. When that tax falls short, the difference is borrowed from the federal general fund. So the shortfall comes from the public, and not from the extraction that severed the cost. Reclamation bonds posted under the 1977 law fall several billion dollars short of the reclamation they are supposed to guarantee. And no Foreclosure Bond of any kind stands against the obligation that Social Security carries from one generation to the next.

This is the accountability gap, and it is a measurable fact, not a figure of speech. The bond that has been posted is a small fraction of the cost that has been severed. Chapter 6 supplies the method for measuring the gap, and Chapter 9 takes up the policy of closing it.

What is owed is a separate question from what any society is willing or able to collect today. Keeping the two apart guards against a misreading that this whole argument invites. How to get from today's nearly empty bonds to the real value of the commons is a political question. Chapter 9 treats it as an open one, and it does not pretend to legislate a single answer. The method applies whether or not anyone is ready to post the bond. What it contributes is accounting, not advocacy. Whoever holds power, and whatever they are willing to do, an honest count gives the same answer to the question of how large the bond would have to be.

The family, again

Come back, finally, to the room where this chapter began. A pacemaker kept a heart beating through years of decline. At seventy-seven a woman kept watch more than eighty hours a week, and no one kept a ledger of it. Butler's father died in the spring of 2008 with her mother at his

side. The family's request to have the pacemaker turned off had never been granted. "Inside his chest, his pacemaker was still quietly pulsing."

Nothing in that room has changed since the first page. What has changed is that you have now seen the same thing happen in every ring around it. The room is not a comparison for the other cases. It is the same mechanism at its smallest size, and in that room the person left holding the bill was Butler's mother.

The rings end where they began, with family. The people who will hold the largest share of the bill are our own descendants, and they haven't been born yet. They are farther away than Baotou, but the distance is in time and not in space, and they cannot object to anything that is decided now. The cost of what is extracted today can be pushed onto them, as a habitable atmosphere is spent, irreplaceable materials are drawn down and options are closed off. It is the same severed cost as all the others. The only difference is that the people who pay live in a later time, and no one can file a claim from the future against the past. In the case of the atmosphere, the person left holding the bill is everyone who will ever be born.

So the question this chapter leaves is not whether severed costs are real. Every ring settles that. It is the question a family would ask on behalf of its own children. What would it take to post the bond before the cost is severed, rather than chasing it after the beneficiary has gone?